Context-Aware Image Descriptions for Web Accessibility

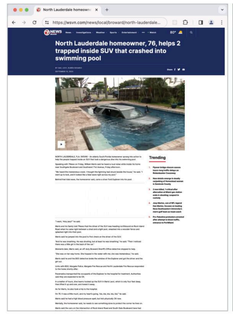
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The University of Texas at Austin Computer Science College of Natural Sciences

Why Context Matters for Image Descriptions

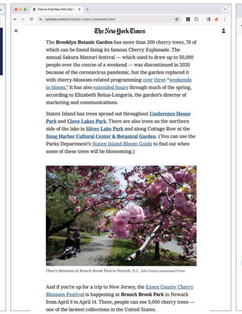
Images are important for understanding information on the web, but often images are missing alternative text such that they are inaccessible to blind and low vision (BLV) people who use screen readers to access the web.

News



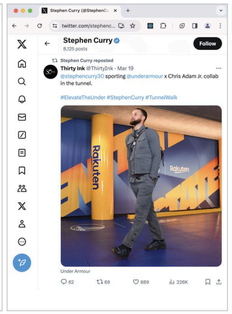
alt = “”

Travel



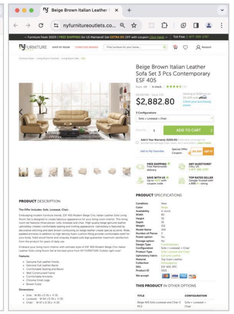
alt = “A close-up of a pink cherry tree.”

Social Media



alt = “Image”

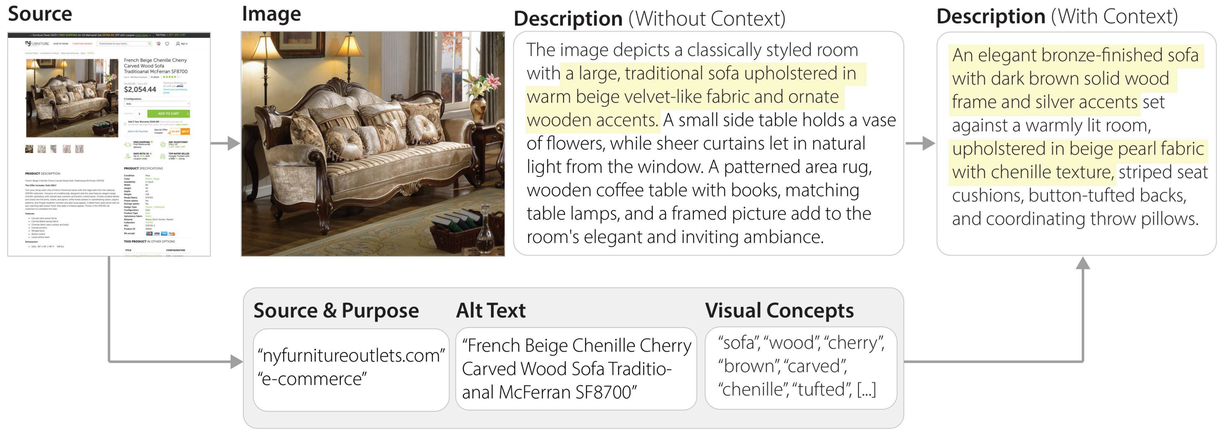
Shopping



alt = “Beige Brown Italian Leather Sofa Set 3 Pcs Contemporary ESF 405”

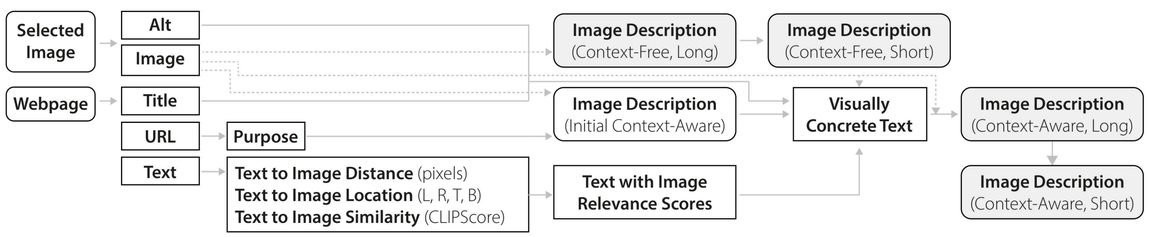
Context-Aware Image Descriptions

Our system provides context-aware descriptions for images by considering the image along with extracted details from the image source to craft a description that uses correct visual terminology (e.g., chenille texture rather than velvet) and focuses on the relevant item (e.g., the sofa rather than the room).



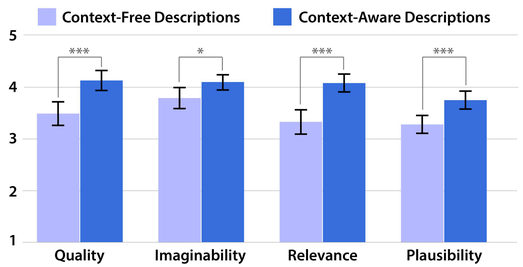
Prototype System

The system takes a webpage and a selected webpage image as selected by the user then provides context-aware descriptions based on both the webpage content and selected image. In addition to the image itself, the descriptions take into account specific visual concepts (e.g., specific places, people, and objects) and their layout (e.g., size, position with relation to the image) along with the overall source and purpose of the webpage.



Results

Participants of the User Study preferred the context-aware image descriptions generated by our tool for Overall Quality, Relevance, Imaginability, and Plausibility in comparison to the context-free image descriptions generated by GPT-4V.



BLV Participants Ratings for Overall Quality, Relevance, Imaginability, and Plausibility in Task 1

